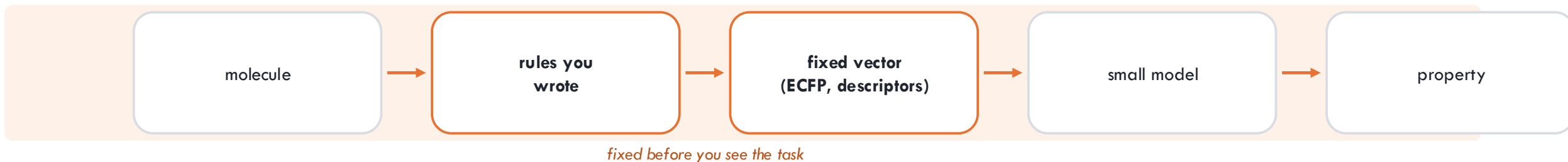


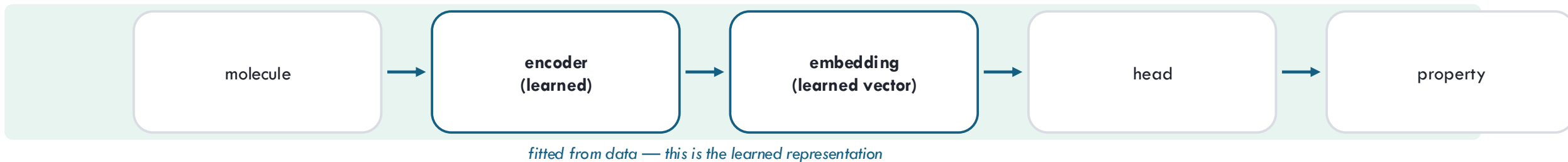
Learned Representations

Letting the model choose the features

YOU choose the features



THE MODEL chooses the features



Plan

- **Why learn a representation?** the label bottleneck (10 min)
- **Unsupervised learning:** clustering, chemical space maps, what they do and do not tell you (20 min)
- **Autoencoders and VAEs:** compression, latent spaces, and an honest look at molecular generation (20 min)
- — break —
- **Self-supervised pre-training:** masked prediction and contrastive learning (15 min)
- **Transfer learning:** frozen, partial, full — and where it genuinely pays (15 min)

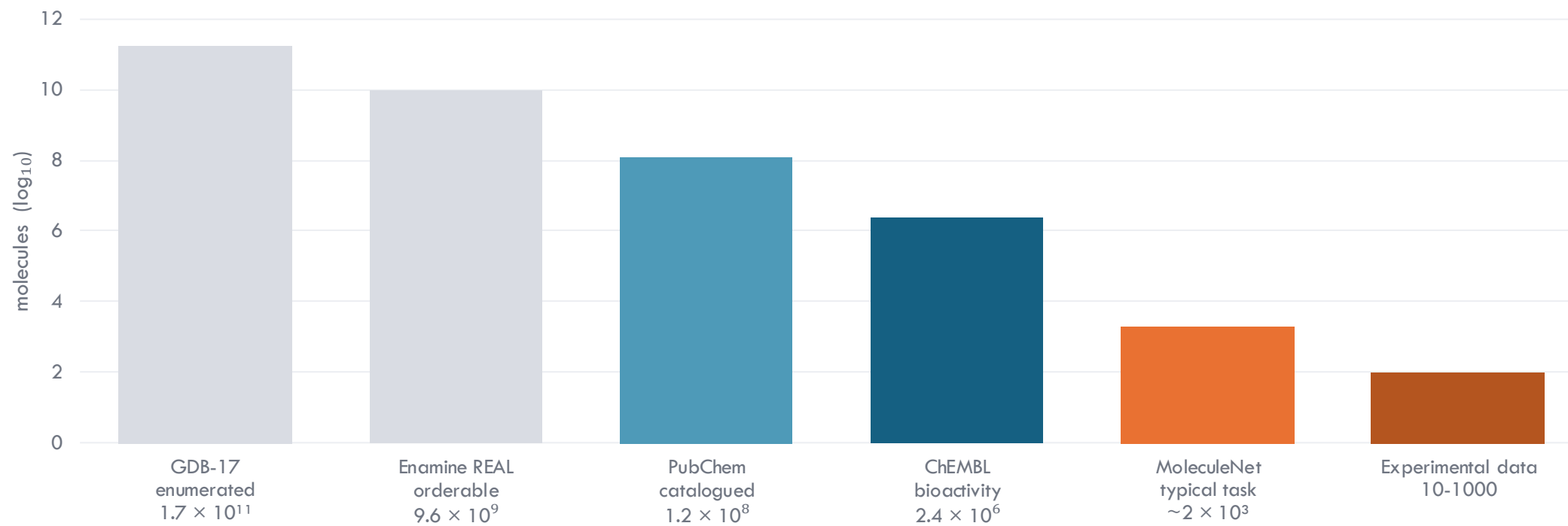
Tutorial at 15:15 — you will build the map and measure the crossover yourself, on real data.

01

Why learn a representation?

- Hand-crafted vs learned features
- The label bottleneck
- Three kinds of supervision

Labels are expensive not inputs



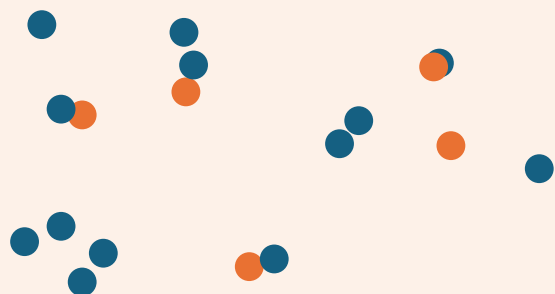
Labels are the scarce resource

GDB-17: Ruddigkeit et al., J. Chem. Inf. Model. 2012, 52, 2864 · Enamine REAL (2026) · PubChem · ChEMBL

Learning modes

SUPERVISED

learn $x \rightarrow y$

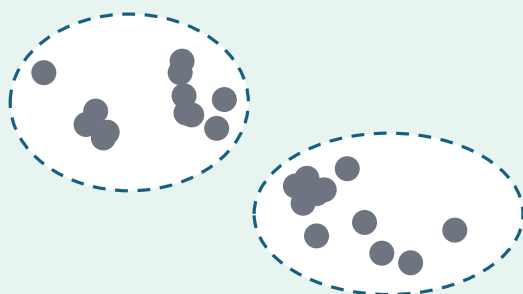


every point carries a label

labels are expensive

UNSUPERVISED

learn the shape of x

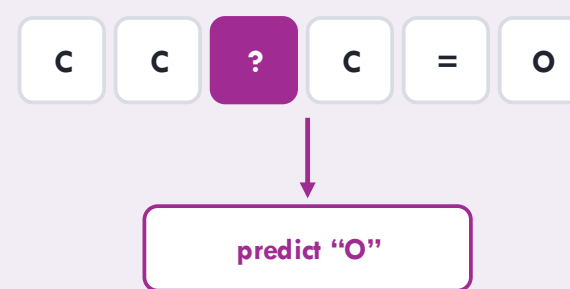


no labels

unlimited data

SELF-SUPERVISED

learn $x \rightarrow \text{part of } x$



automated labels generation

supervised signal, no annotation

02

Unsupervised learning

- Clustering, dimensionality reduction, density
- Chemical space maps — and how to read them honestly
- VAEs treated separately

Unsupervised learning

1 · CLUSTERING

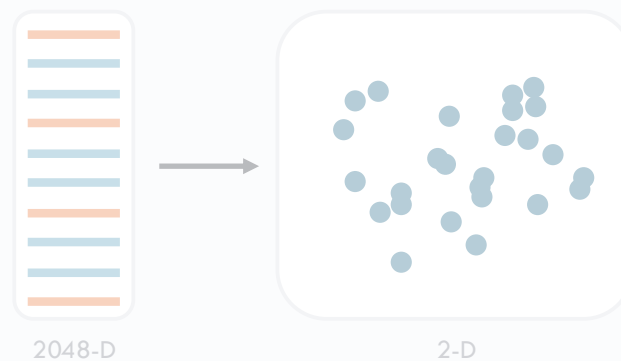
which molecules belong together?



groups, no coordinates

2 · DIMENSIONALITY REDUCTION

can I see 2048 dimensions?



2048-D

2-D

coordinates you can plot

3 · DENSITY & GENERATION

what does a plausible molecule look like?



sample new points from the model

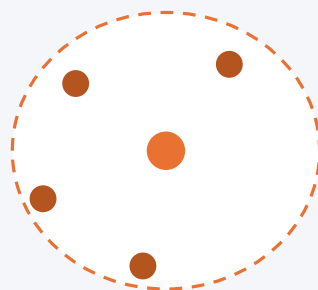
Butina clustering: the cheminformatics default

1 · count neighbours within t



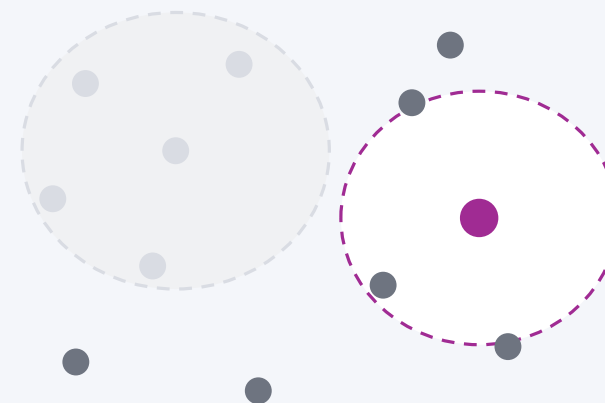
t = Tanimoto cut-off

2 · densest molecule claims its sphere



cluster centre + members

3 · repeat on what is left

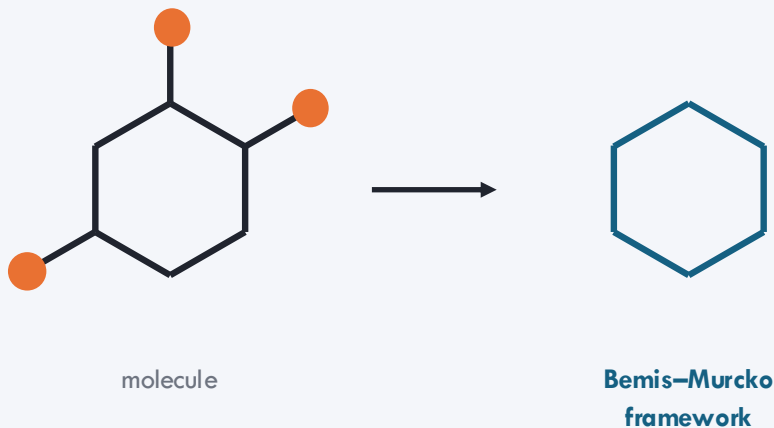


greedy, deterministic, no k to choose

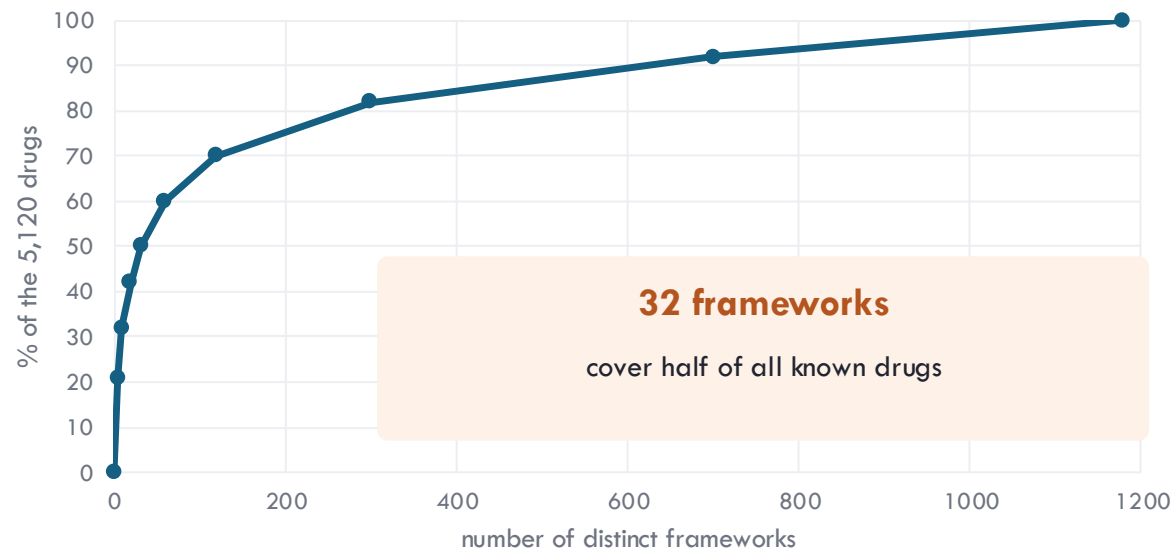
No k to choose, deterministic. Run it at two or three cut-offs and check your conclusion survives.

Scaffolds: chemistry is far more clustered than it looks

strip side chains → keep the framework



scaffold split = hold out whole frameworks,
not random molecules



1,179 distinct frameworks among 5,120 drugs

32 frameworks cover half of all known drugs.

Unsupervised learning

1 · CLUSTERING

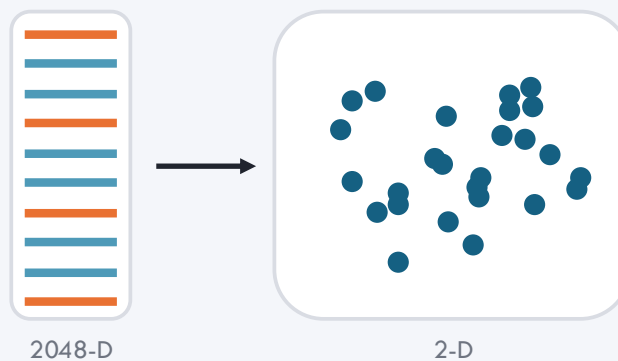
which molecules belong together?



groups, no coordinates

2 · DIMENSIONALITY REDUCTION

can I see 2048 dimensions?



coordinates you can plot

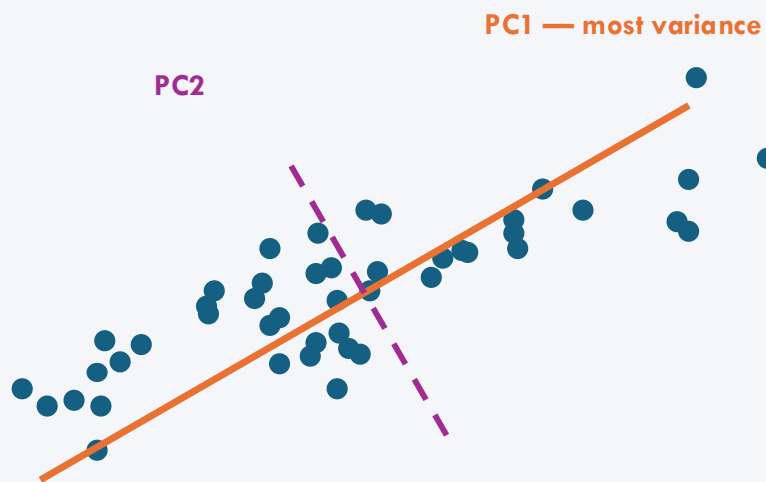
3 · DENSITY & GENERATION

what does a plausible molecule look like?

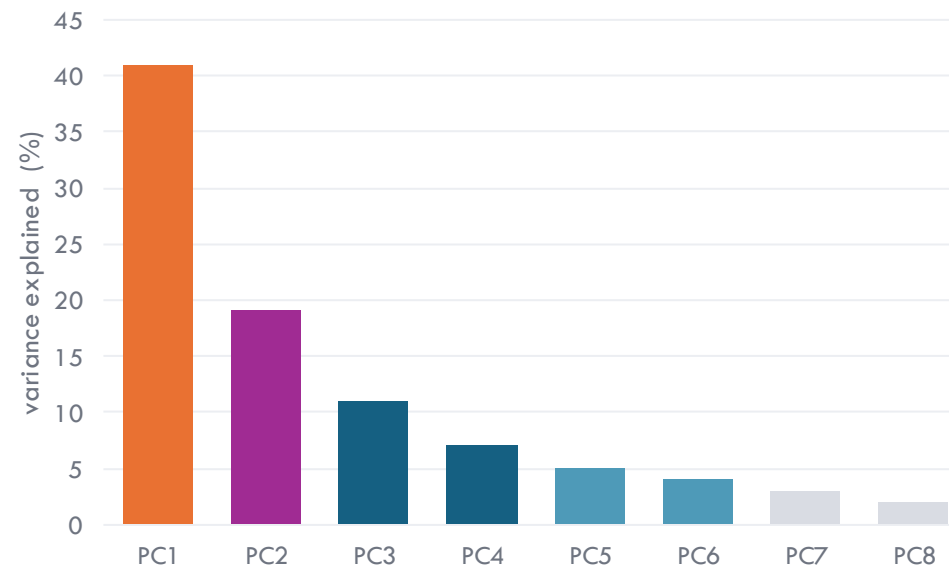


sample new points from the model

PCA: a rotation, and an honest one

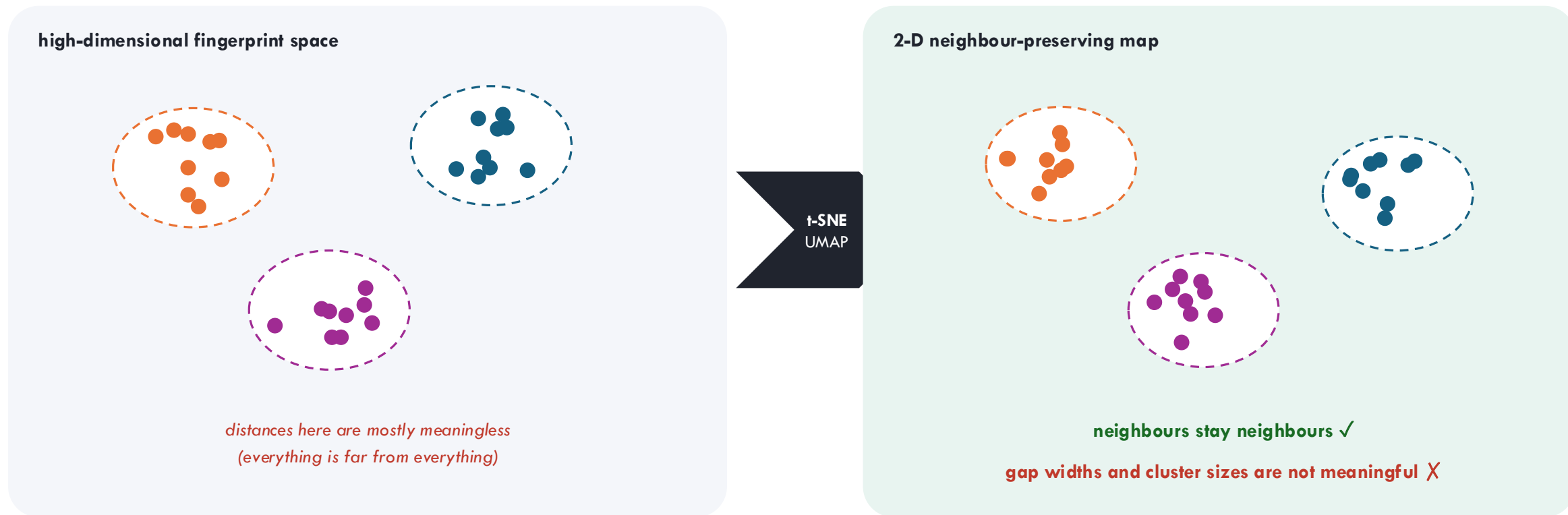


a rotation, nothing more



2 components $\approx$ 60% of the variance — the rest is not noise, it is chemistry

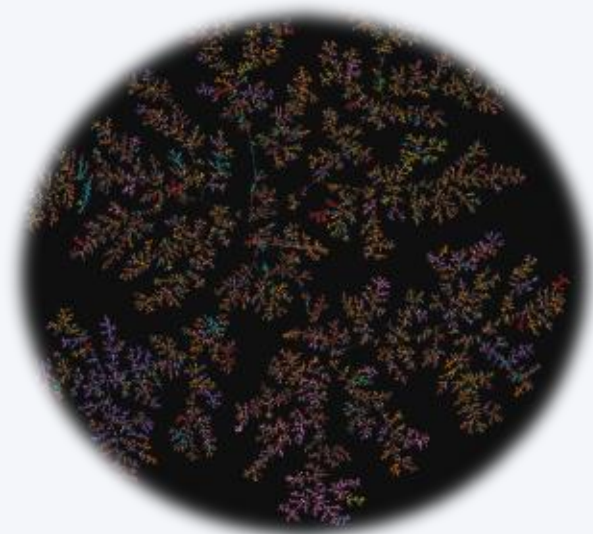
t-SNE and UMAP: keep the neighbors, lose the distances



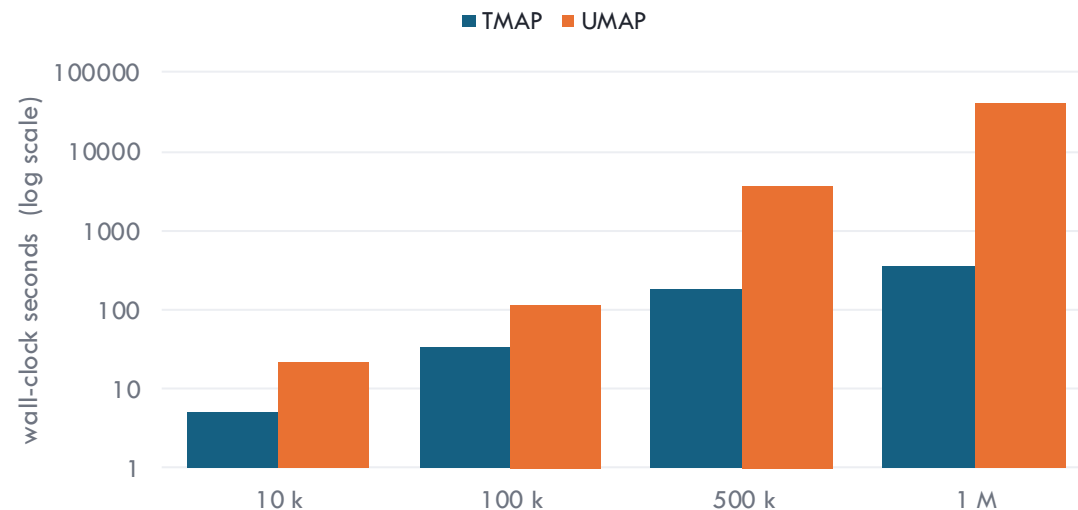
The trade is deliberate: **neighborhoods are preserved, distances are not.**

Algorithm matters at large scale

TMAP: lay out a minimum spanning tree, not a scatter



branches = series of analogues · no overplotting at 10^6 molecules

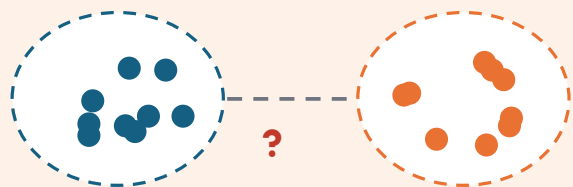


1 M molecules: TMAP 6 min vs UMAP 11.5 h (~117×)

TMAP: ~117× faster than UMAP at 10^6 molecules, and reproducible.

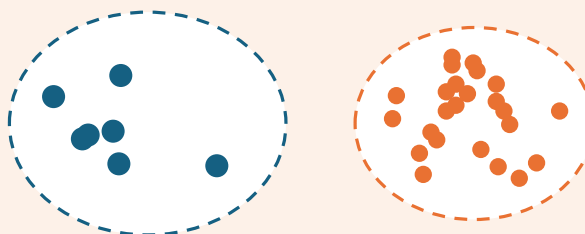
Three things your map is not telling you

✗ gap width means nothing



perplexity changes it

✗ cluster size means nothing

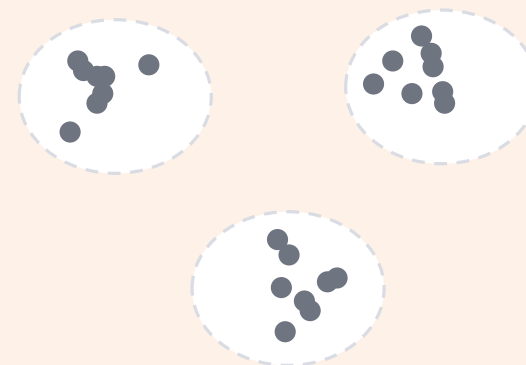


7 molecules

240 molecules

drawn almost the same size

✗ clusters appear in pure noise



input was uniform random

Never read clusters off a map. Cluster explicitly in the original space, then use the map only to display the result.

Unsupervised learning

1 · CLUSTERING

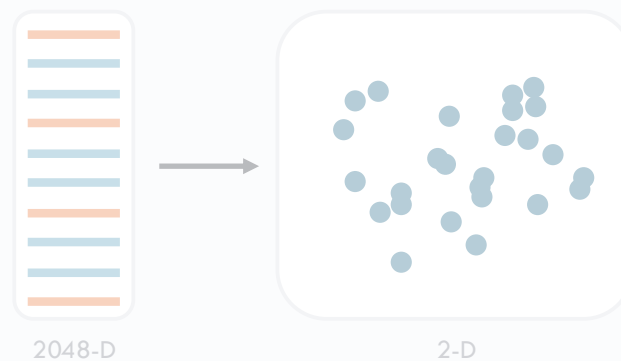
which molecules belong together?



groups, no coordinates

2 · DIMENSIONALITY REDUCTION

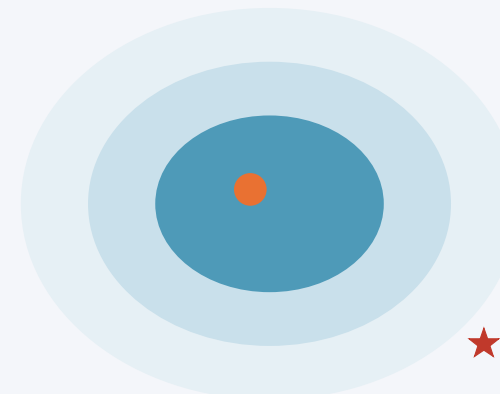
can I see 2048 dimensions?



coordinates you can plot

3 · DENSITY & GENERATION

what does a plausible molecule look like?



sample new points from the model

Case study: the periodic table, from text alone

no chemistry was taught

“... LiFePO_4 cathodes show ...”

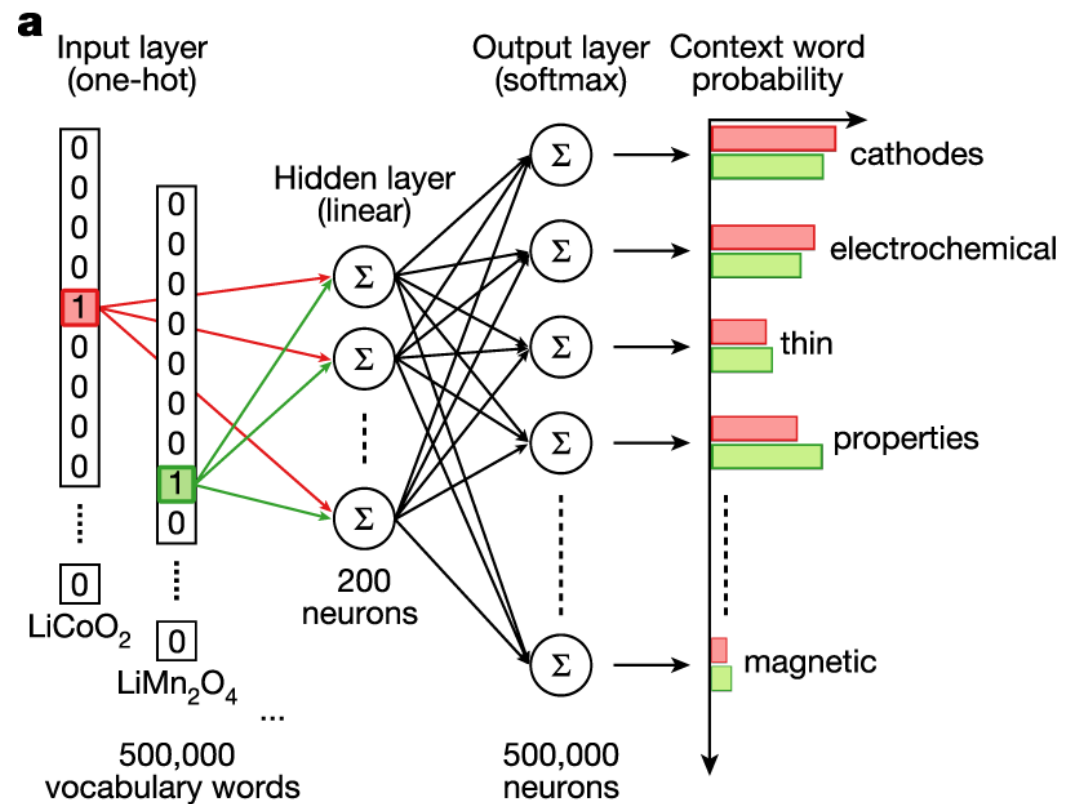
“... we anneal the CoSb_3 skutterudite ...”

“... Bi_2Te_3 thermoelectric performance ...”

3.3 M materials-science abstracts

1.5 M used · word2vec · 200-D

the only signal is which words co-occur



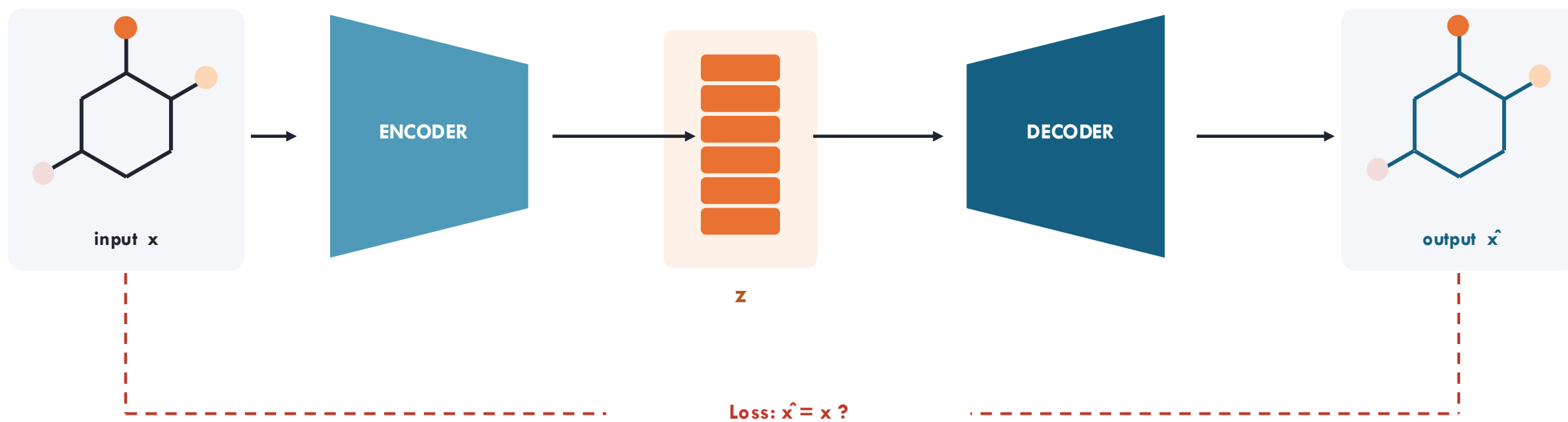
Unsupervised learning on ~1 M abstracts produce a representation.

03

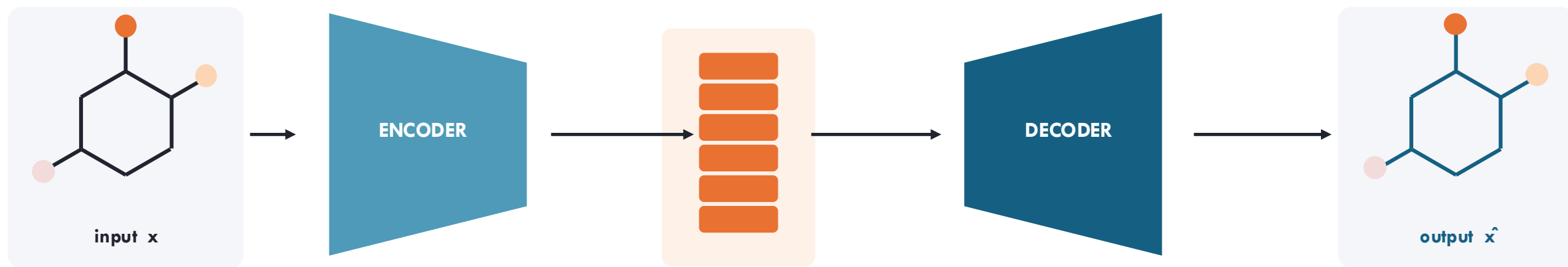
Autoencoders and VAEs

- Compression as representation learning
- Why a plain autoencoder is not enough
- What VAEs are — and are not — good for

The autoencoder: learn by rebuilding the input



Compressing the representation



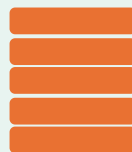
bottleneck too wide

z is as big as x



bottleneck just right

$z \ll x$

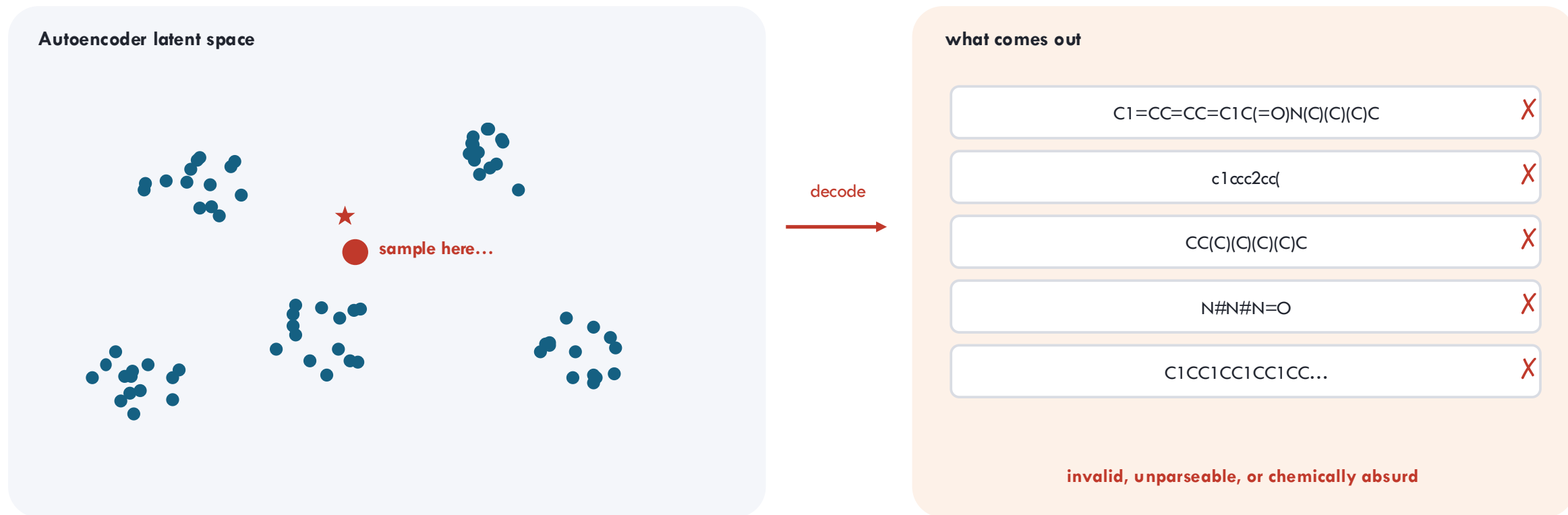


bottleneck too narrow

$z \approx 2$



The problem: a plain autoencoder learns islands



The VAE in one idea: encode a cloud, not a point

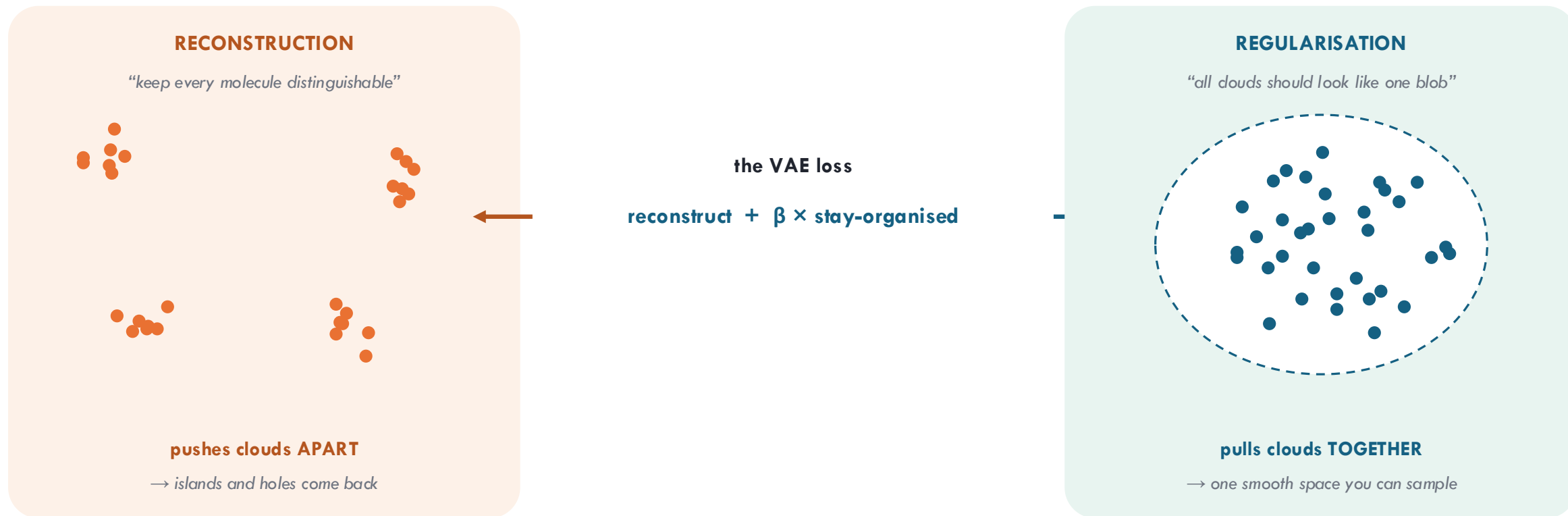
Autoencoder (AE): $x \rightarrow \text{one point}$



VAE: $x \rightarrow \text{a small cloud}$

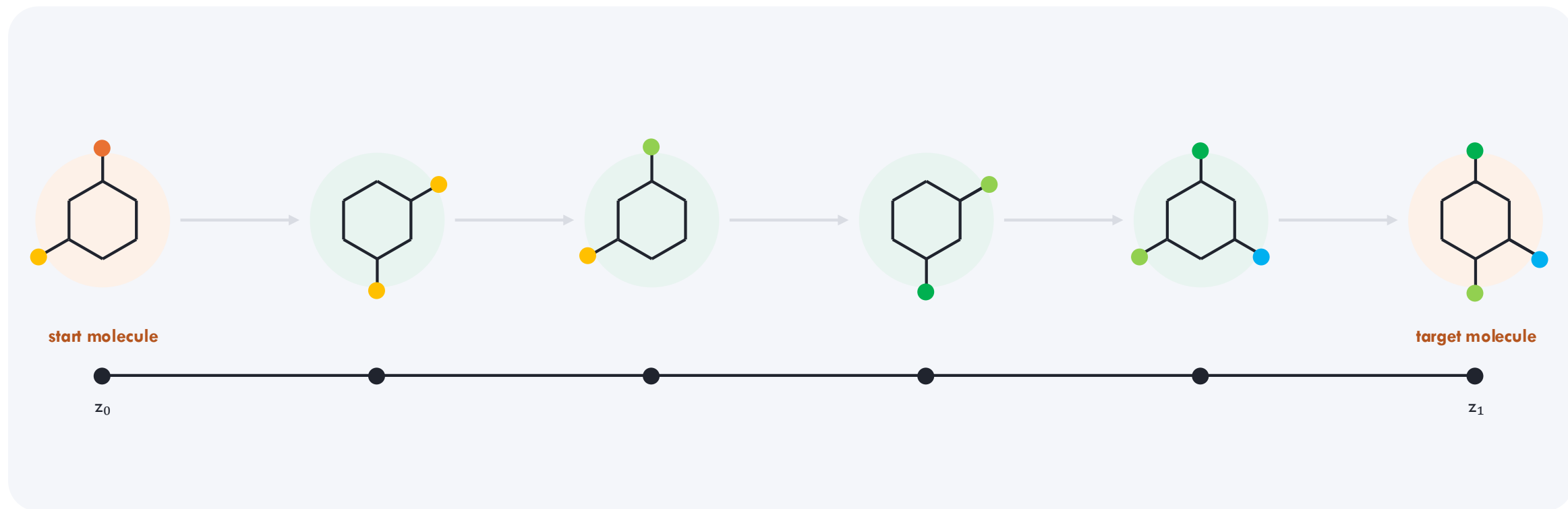


Two forces, and a dial between them

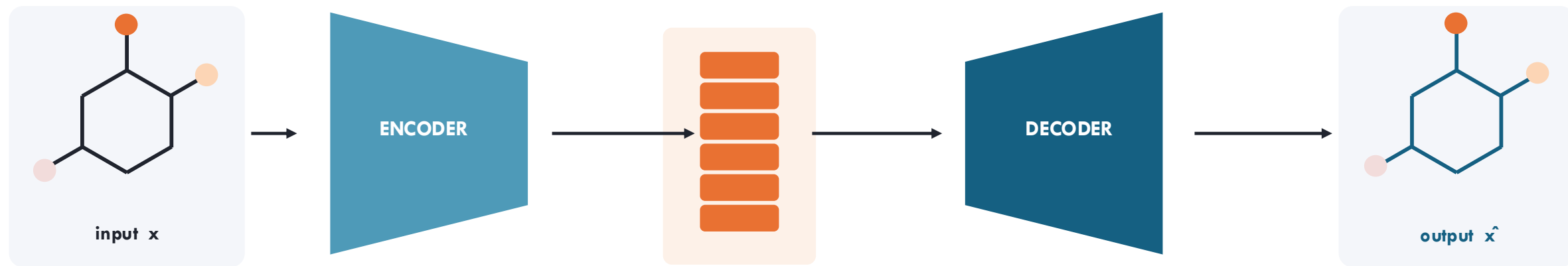


β is the whole design decision. Too small and you are back to islands; too large and the decoder ignores z entirely.

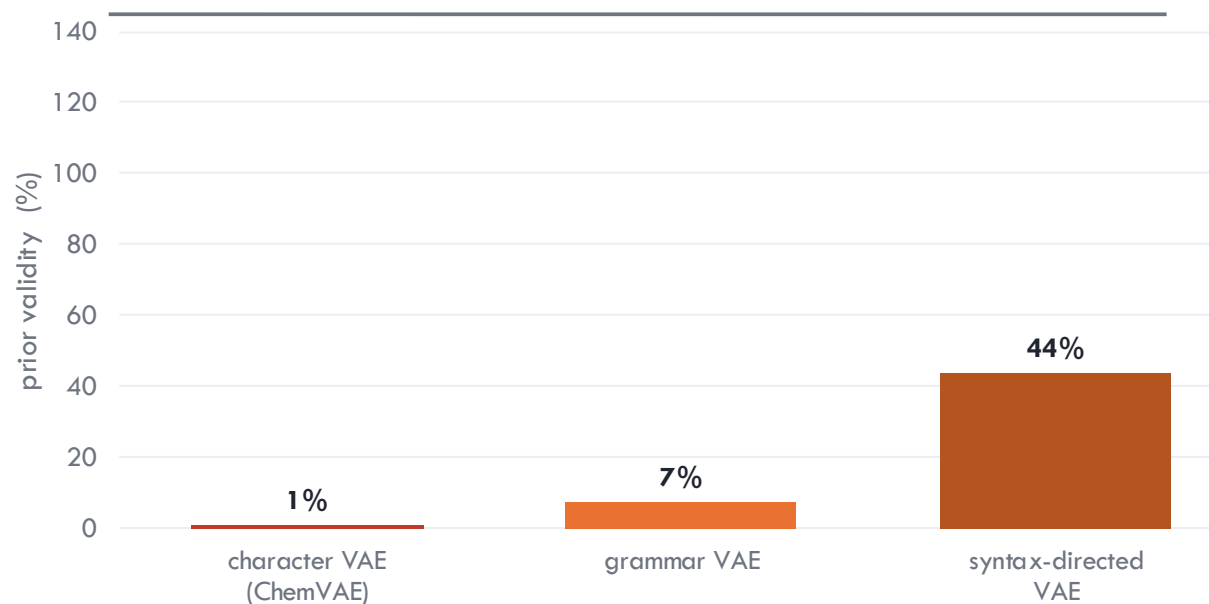
Attempt to build a "continuous" molecular space



Solving the Validity Issue



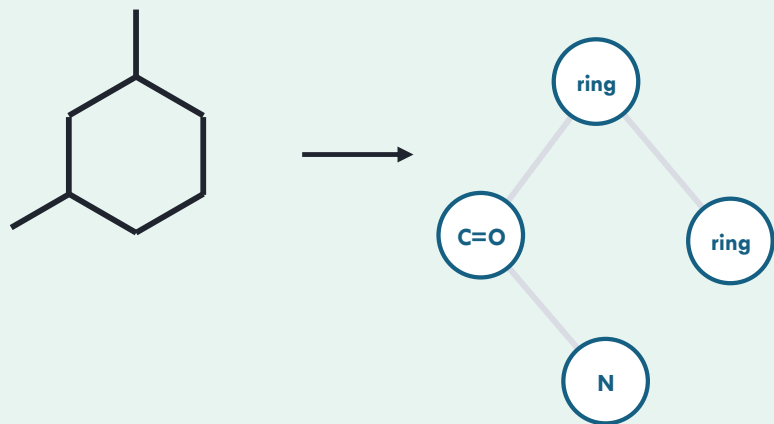
learned



Jin, Barzilay & Jaakkola, ICML 2018, Table 1 · Krenn et al., Mach. Learn.: Sci. Technol. 2020, 1, 045024

Enforcing Validity

JT-VAE — assemble from a vocabulary of valid pieces

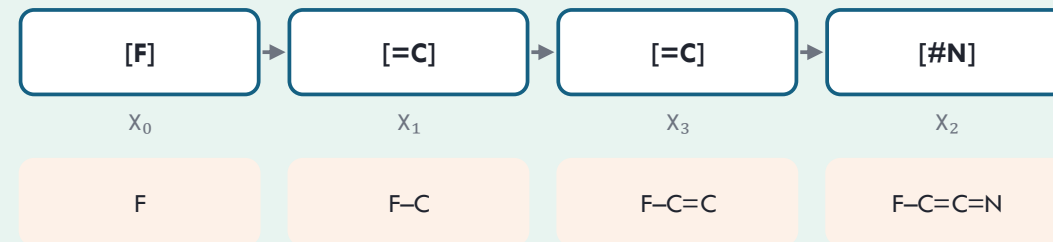


junction tree of substructures

780-piece vocabulary · illegal joins are masked out during decoding

100 % valid

SELFIES — a grammar that cannot fail



asked for a triple bond, valence allowed only a double → clamped, never rejected

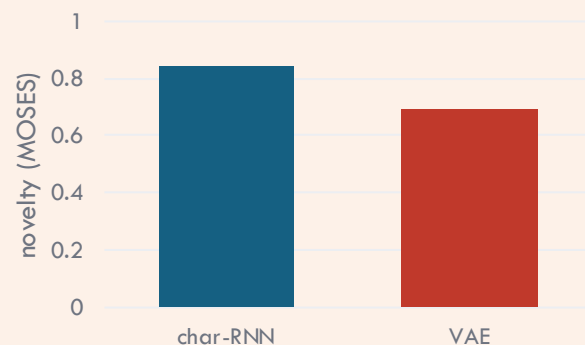
100 % valid · but only local valence is enforced —
not aromaticity, not ring strain, not stability

1 random mutation: SMILES 9.9 % → SELFIES 100 % valid

Both guarantee **syntactic** validity only. A 100%-valid molecule can still be one no chemist would draw.

Validity is not necessarily chemically relevant

it memorises



31 % of what the VAE “designs”
is already in the training set

MOSES benchmark

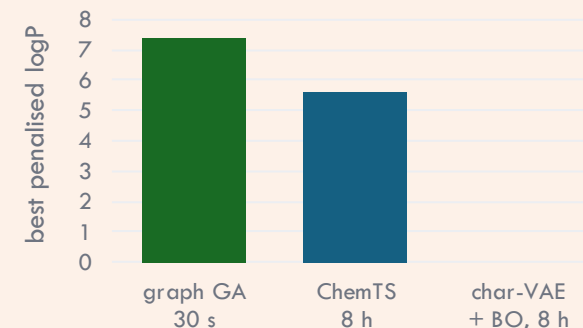
100 % valid $\neq$ good

model	valid %	FCD ↓
Molecule Chef	99.0	0.73
CGVAE	100.0	11.73
char-VAE	12.0	37.65

the model with perfect validity
matches the real distribution worst

FCD = distance to real chemistry

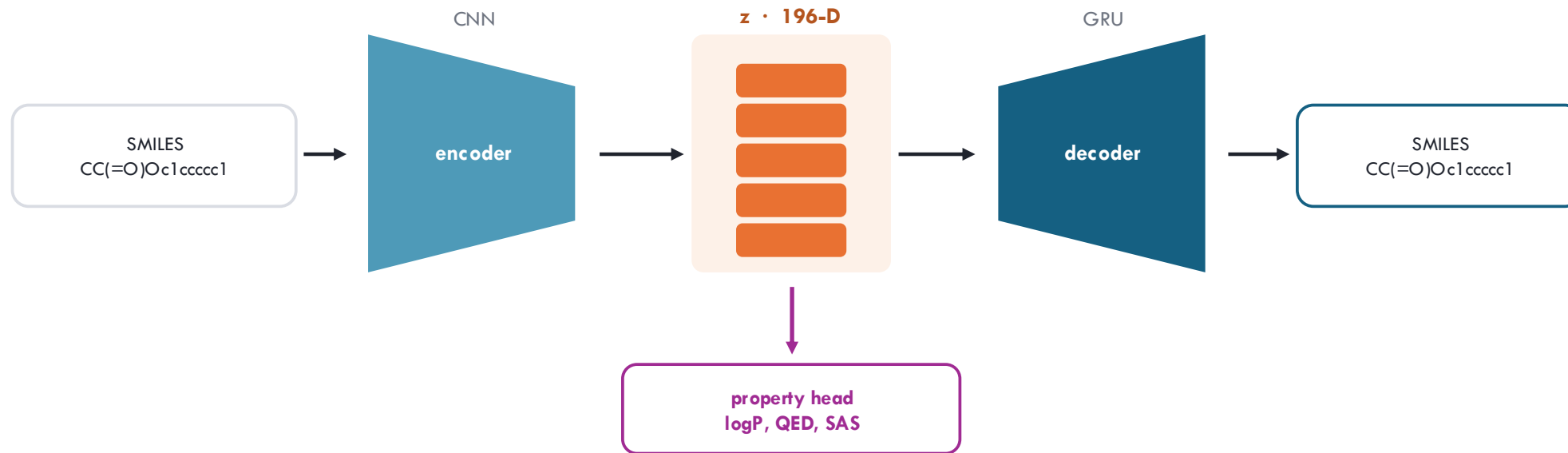
a genetic algorithm wins



30 seconds on a laptop beats
8 hours of VAE + Bayesian opt.

Jensen, Chem. Sci. 2019

ChemVAE, 2018



250,000

ZINC molecules, no labels

196

latent dimensions

~4 %

of random z decode to a valid molecule

Joint property training is the idea

Gómez-Bombarelli et al., ACS Cent. Sci. 2018, 4, 268 — validity figures are version- and definition-dependent; see notes

Keep the encoder, throw away the decoder

as a molecule generator

- ✗ beaten by a graph genetic algorithm
- ✗ beaten by an ordinary SMILES LSTM
- ✗ beaten by random screening of ZINC
- ✗ only ~30 % of proposals are synthesisable

*the field moved to diffusion, GAs and LLMs
(→ Thursday's lecture)*

as a representation learner

- ✓ a fixed-length vector for any molecule
- ✓ trained without a single label
- ✓ a smooth space you can optimise inside
- ✓ the encoder is reusable

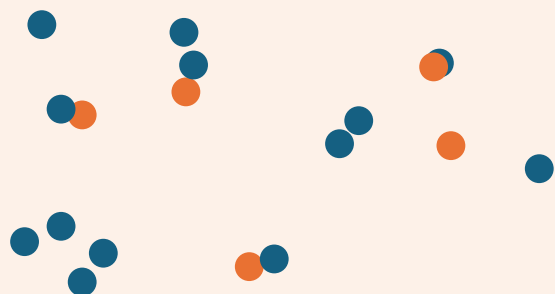
Break

10 minutes — back for pre-training and transfer learning

Learning modes

SUPERVISED

learn $x \rightarrow y$

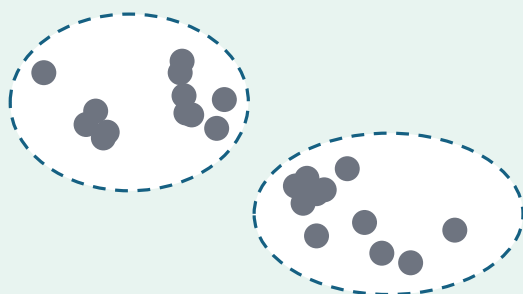


every point carries a label

labels are expensive

UNSUPERVISED

learn the shape of x

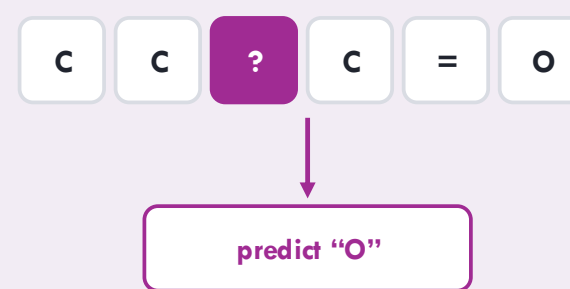


no labels

unlimited data

SELF-SUPERVISED

learn $x \rightarrow \text{part of } x$



automated labels generation

supervised signal, no annotation

04

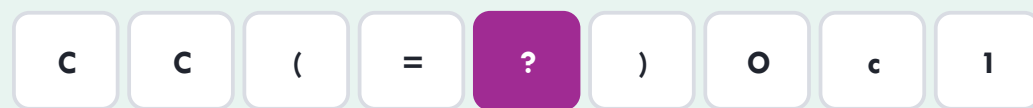
Self-supervised pre-training

- Masked prediction and contrastive learning
- Chemical language models at scale
- Pre-training graph neural networks

Two ways to make the data label itself

A · MASKED PREDICTION

hide part of the molecule, predict it back



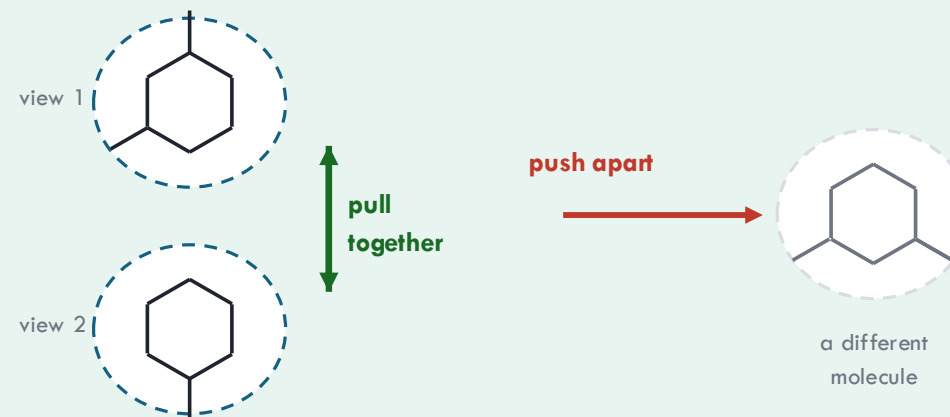
it was "O"

to fill the gap the model must learn valence,
ring closure and functional-group grammar

ChemBERTa · MolFormer · attribute masking on graphs

B · CONTRASTIVE

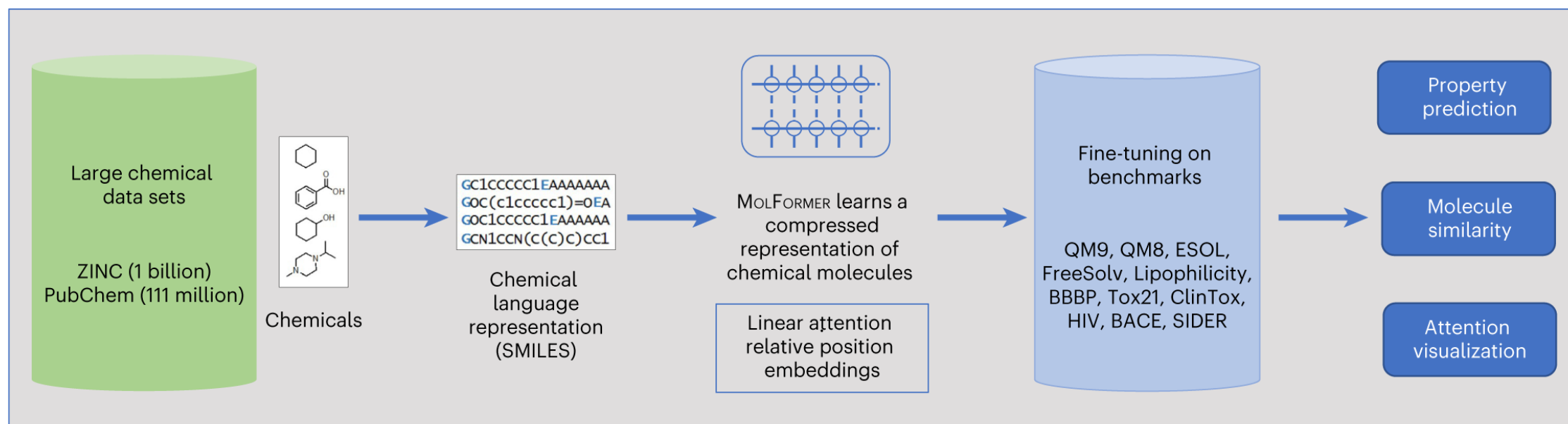
two damaged views of one molecule must agree



MolCLR · GraphCL · InfoMax

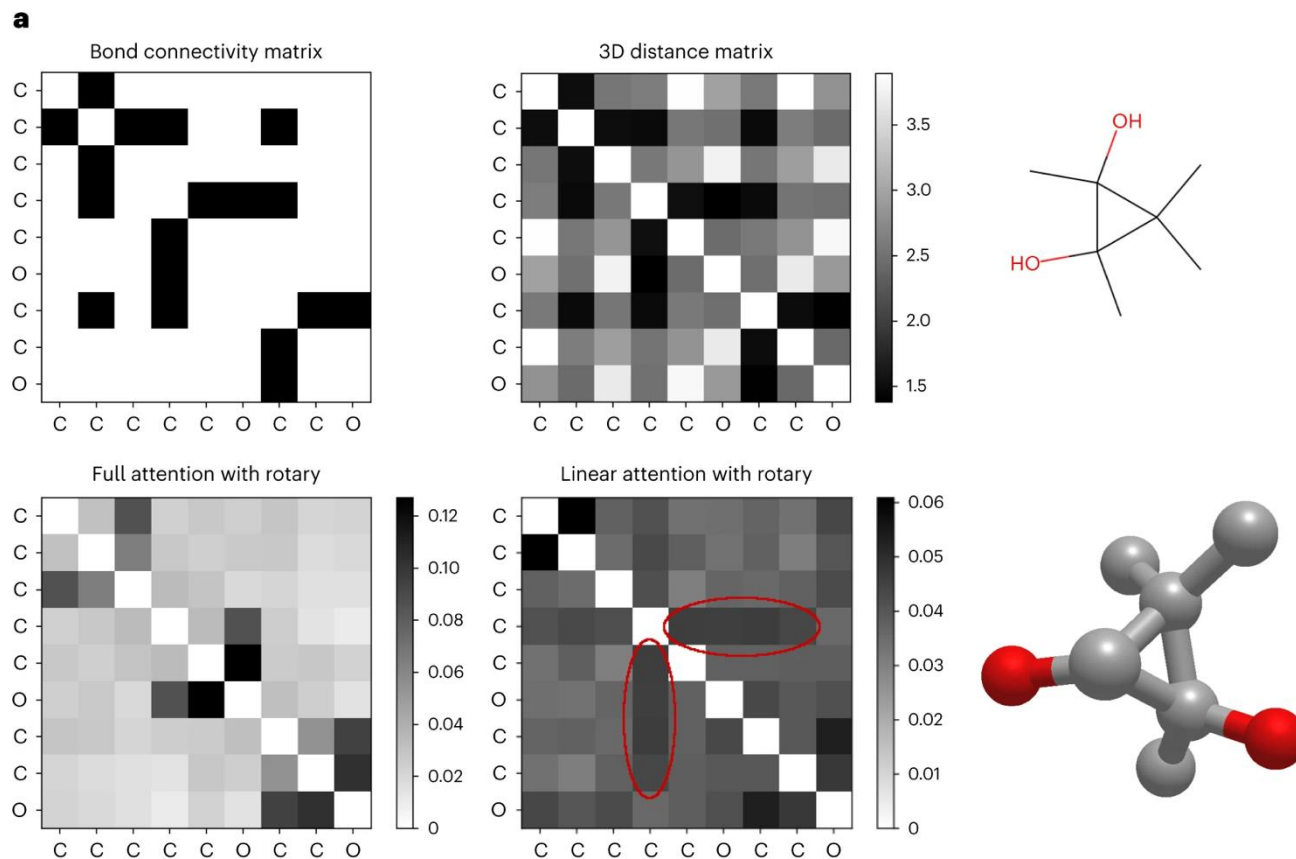
Masked prediction teaches **local chemistry**; contrastive teaches **what to ignore**. Both need only unlabelled molecules.

Pretraining with masked predictions



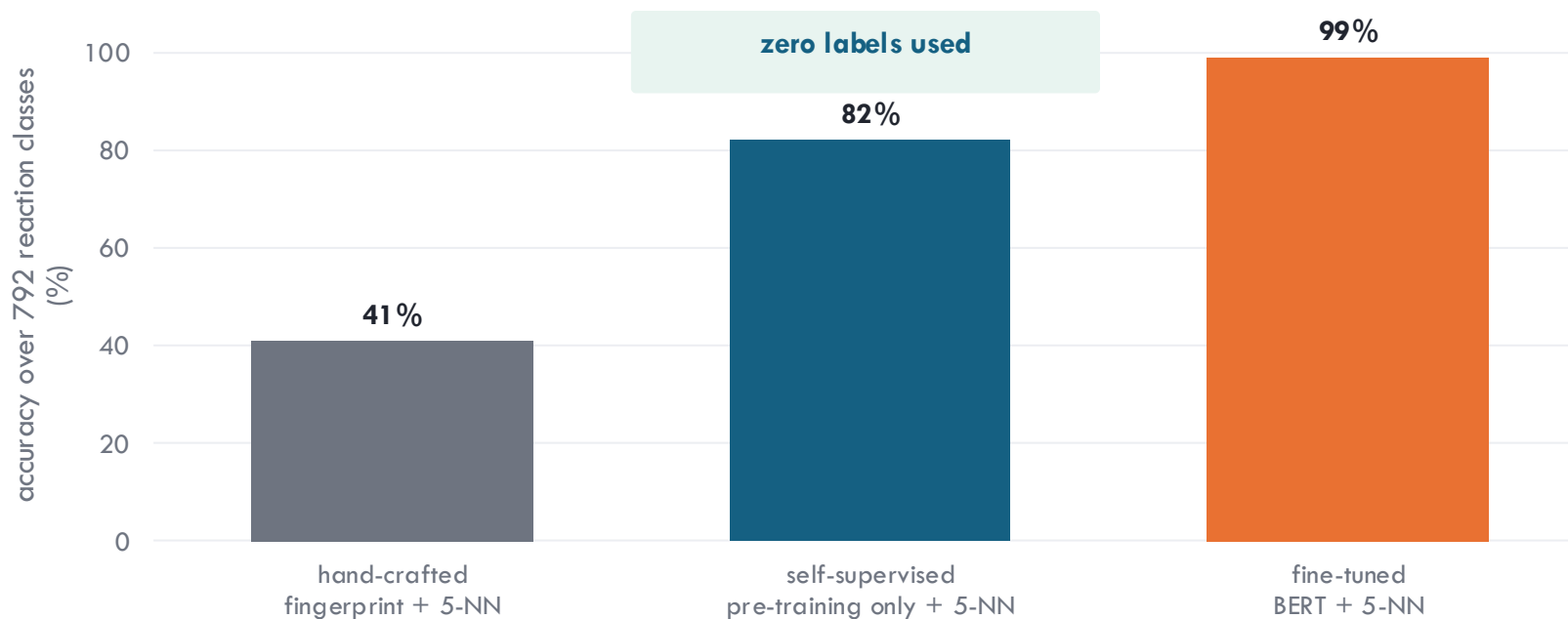
Ross, J., Belgodere, B., Chenthamarakshan, V. et al. Large-scale chemical language representations capture molecular structure and properties. *Nat Mach Intell* 4, 1256–1264 (2022)

What does the pretraining learns?



Ross, J., Belgodere, B., Chenthamarakshan, V. et al. Large-scale chemical language representations capture molecular structure and properties. *Nat Mach Intell* 4, 1256–1264 (2022)

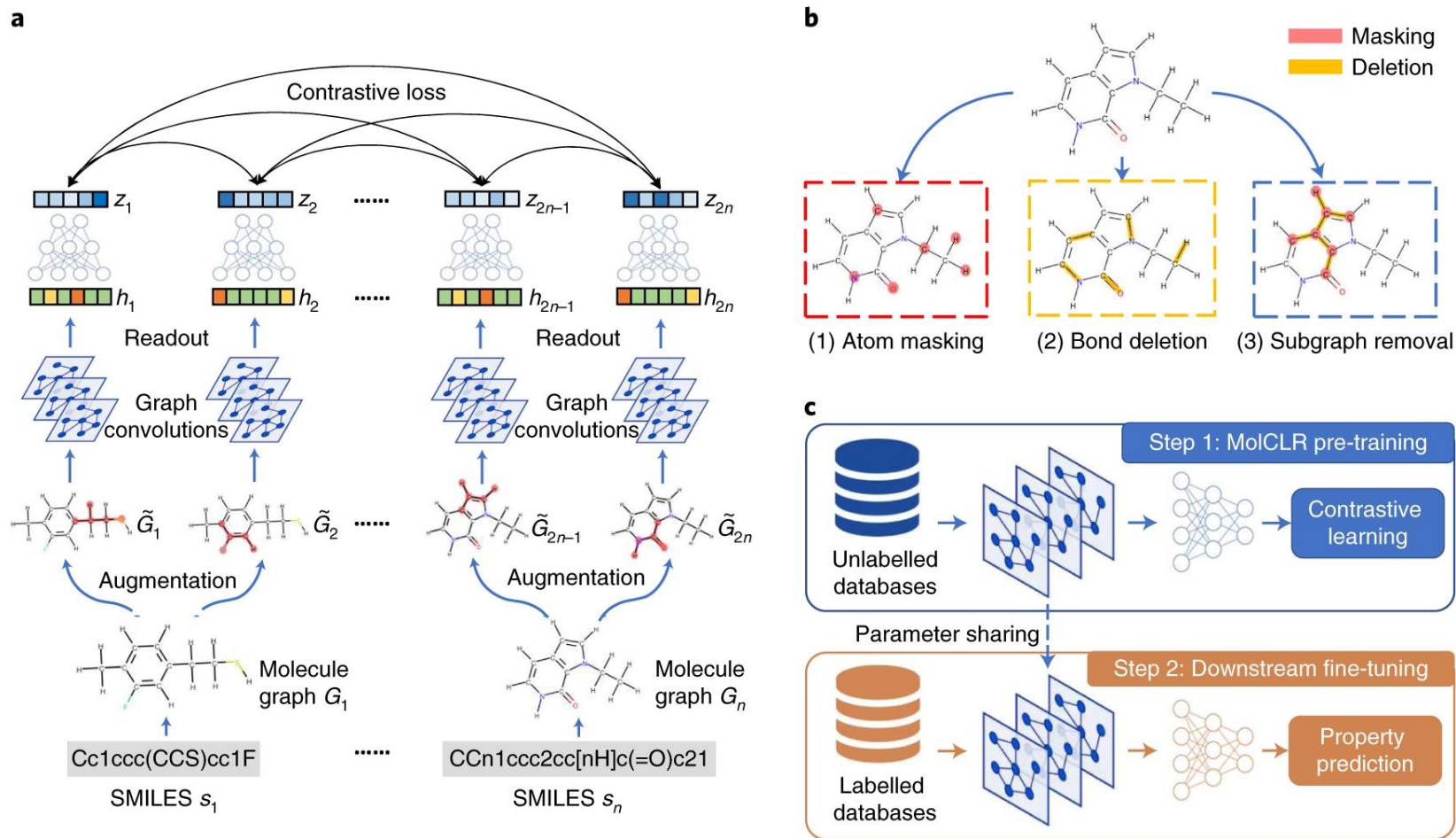
Self supervised learning for reaction



2.6 M unlabelled reactions in · reaction classes come out

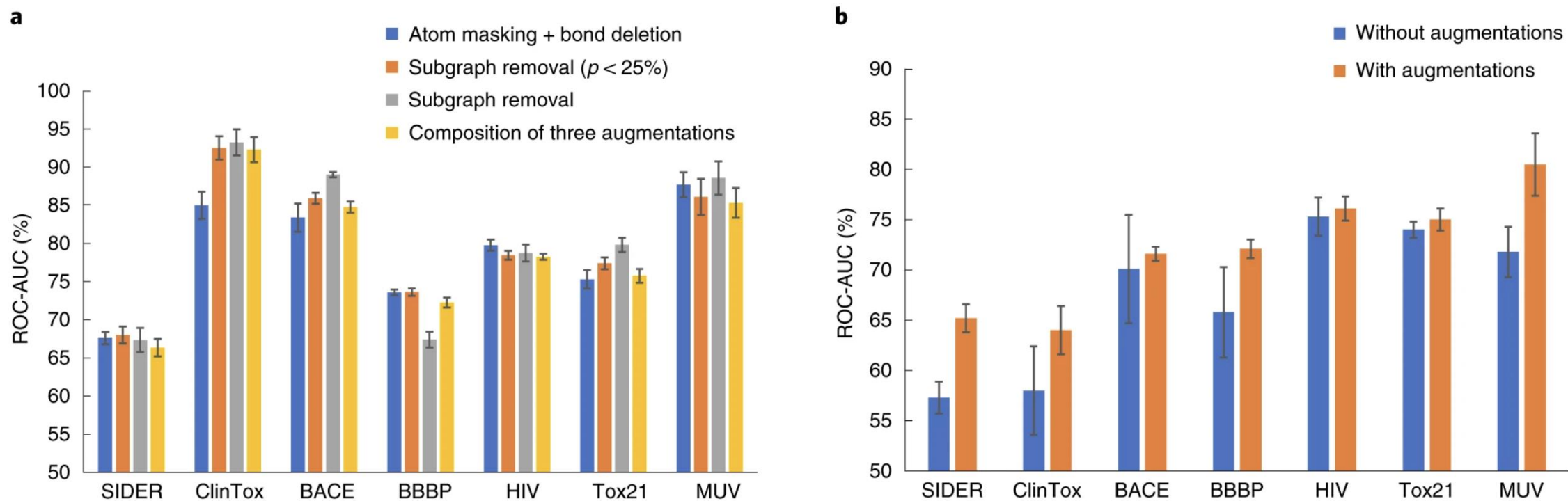
Masked-language pre-training alone recovers most of the reaction taxonomy — 81.9% vs 41.0% for a hand-crafted fingerprint, with zero labels.

Choosing augmentations for contrastive learning



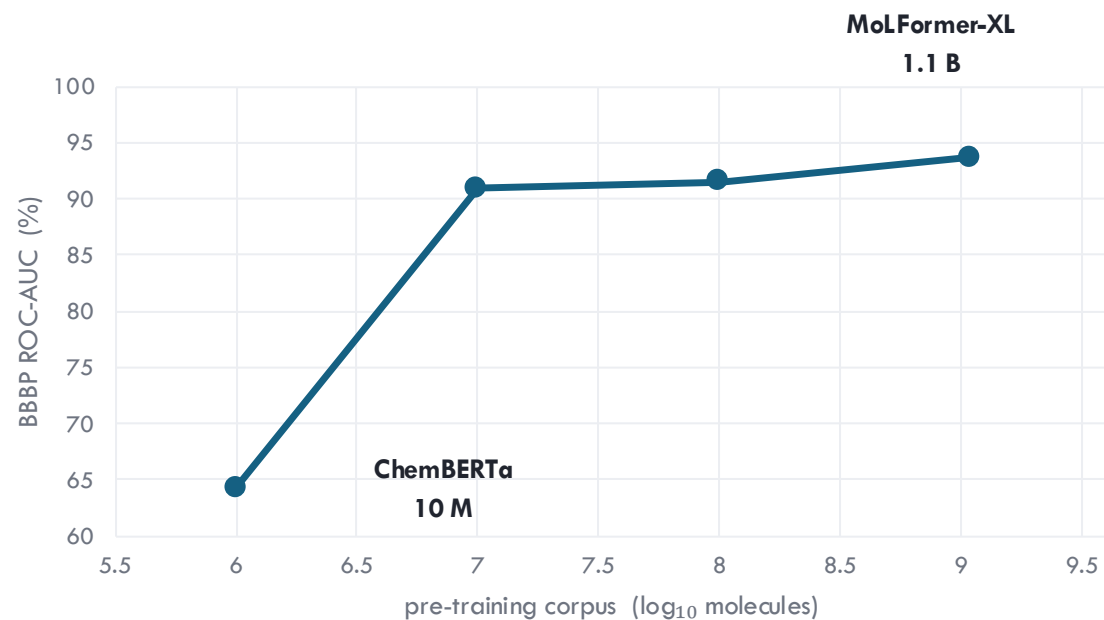
Wang, Y., Wang, J., Cao, Z. et al. Molecular contrastive learning of representations via graph neural networks. *Nat Mach Intell* 4, 279–287 (2022).

Choosing augmentations for contrastive learning

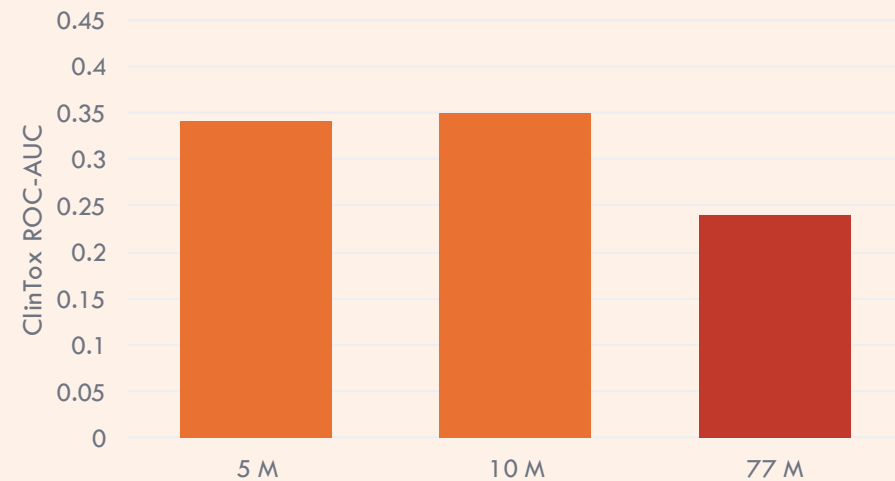


Wang, Y., Wang, J., Cao, Z. et al. Molecular contrastive learning of representations via graph neural networks. *Nat Mach Intell* 4, 279–287 (2022).

Scale and Diversity Impact Pre-training Performance



1.1 B SMILES on 16 GPUs — linear attention made it affordable



ChemBERTa-2, masked-LM: a 15× bigger corpus made this task worse

corpus composition matters more than corpus size

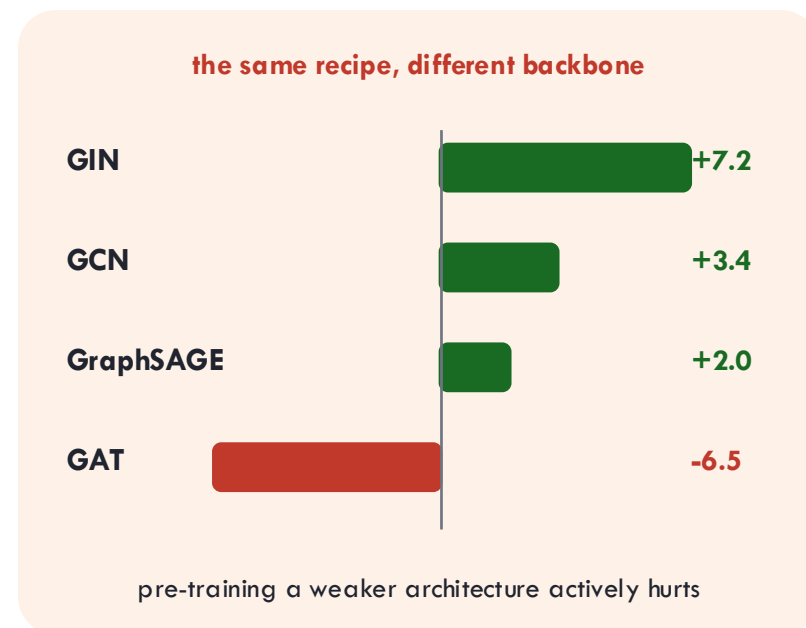
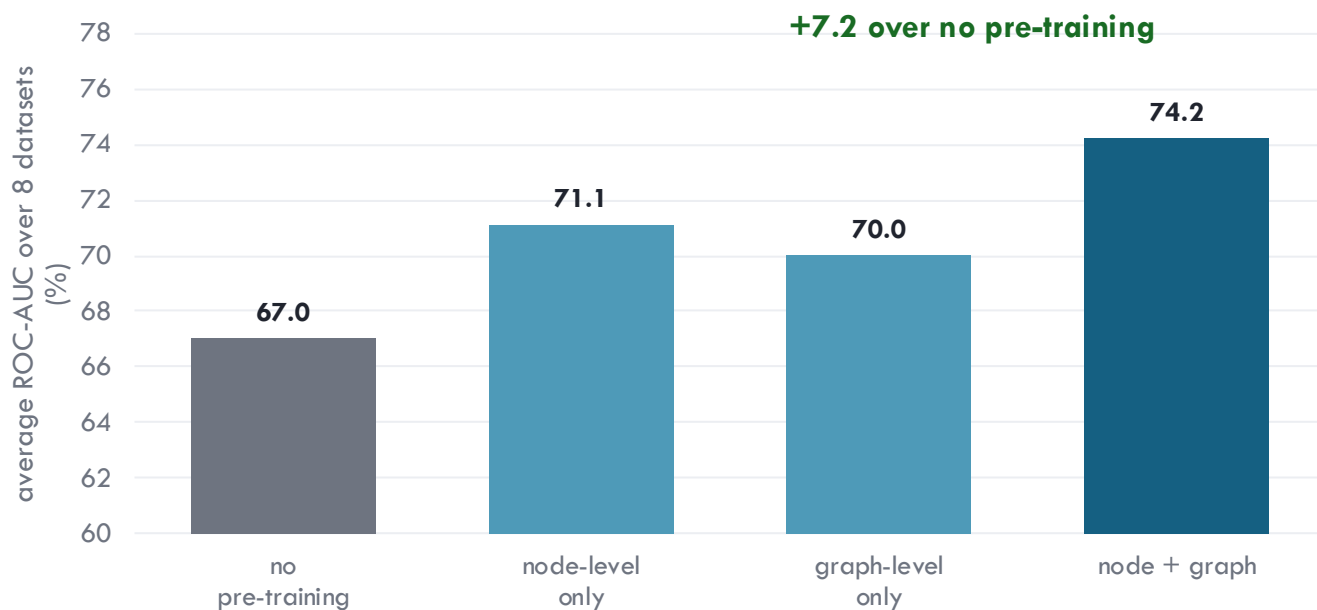
Corpus composition beats corpus size. A 15× bigger corpus made ClinTox substantially worse; a smaller, more diverse one beat a bigger, narrower one.

05

Transfer learning

- Pre-train once, fine-tune many times
- Frozen, partial, full
- When?

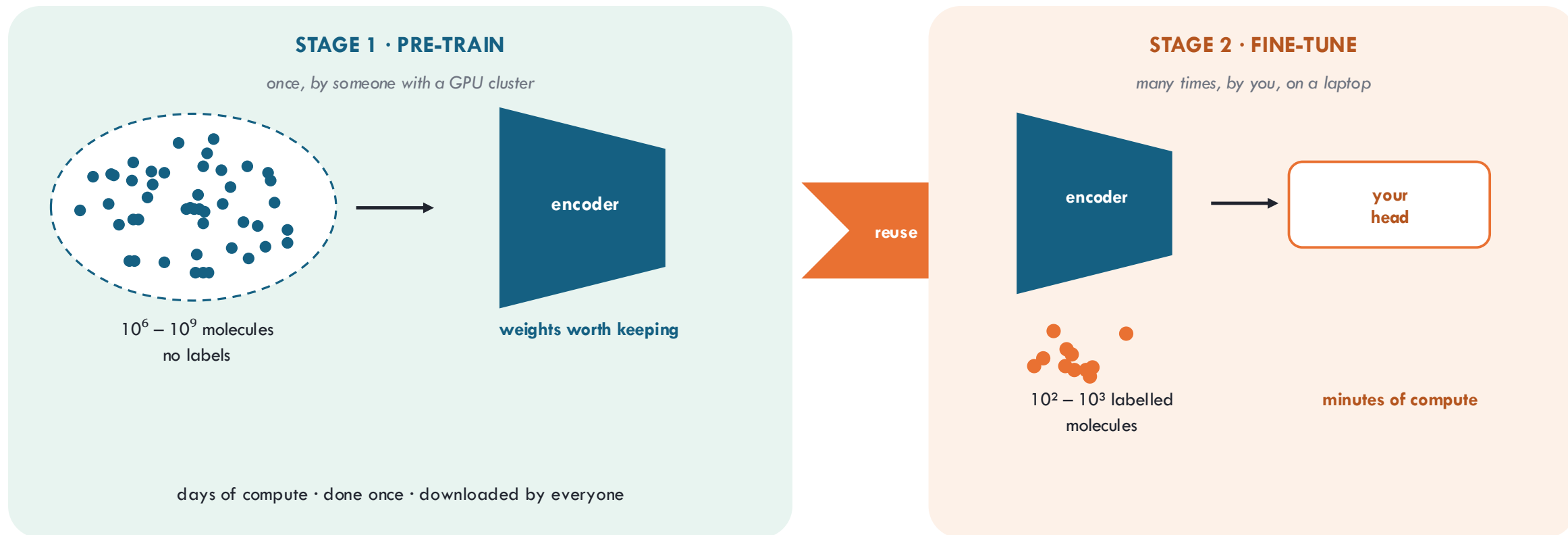
Pre-training GNNs: local and global, or negative transfer



“We pre-trained it” guarantees nothing. **Pre-training a weaker architecture cost 6.5 points of ROC-AUC.**

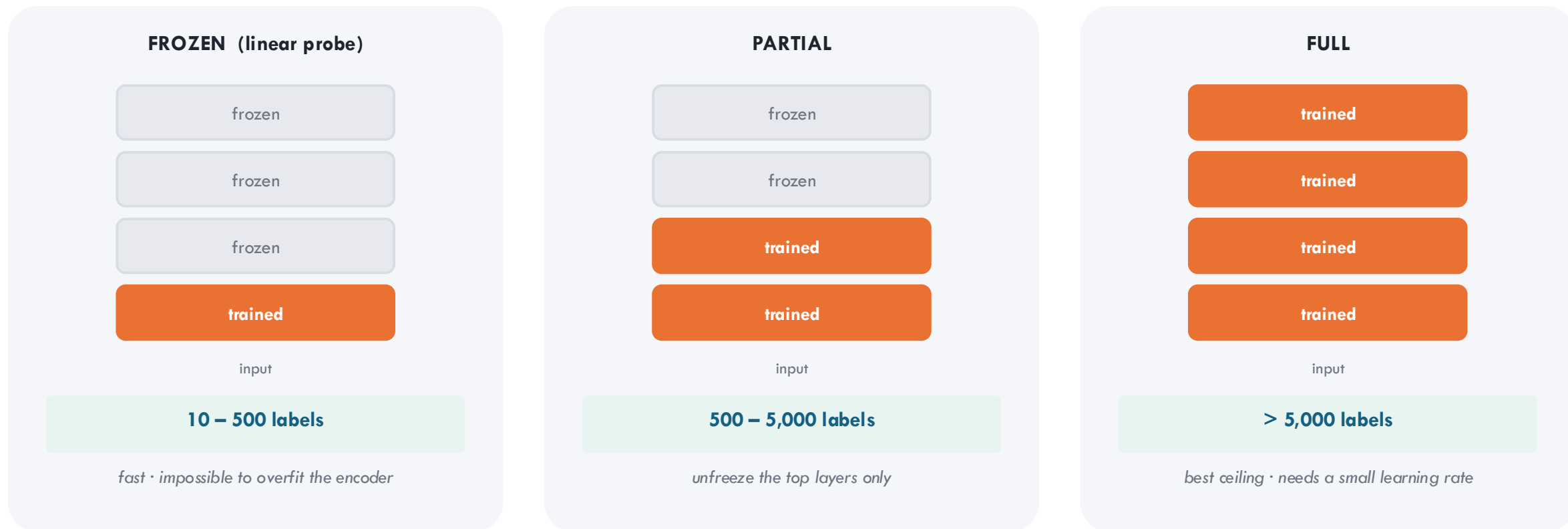
Hu et al., ICLR 2020, arXiv:1905.12265 — chemistry average; the +9.4% in the abstract is the biology result

The recipe: pre-train once, fine-tune many times



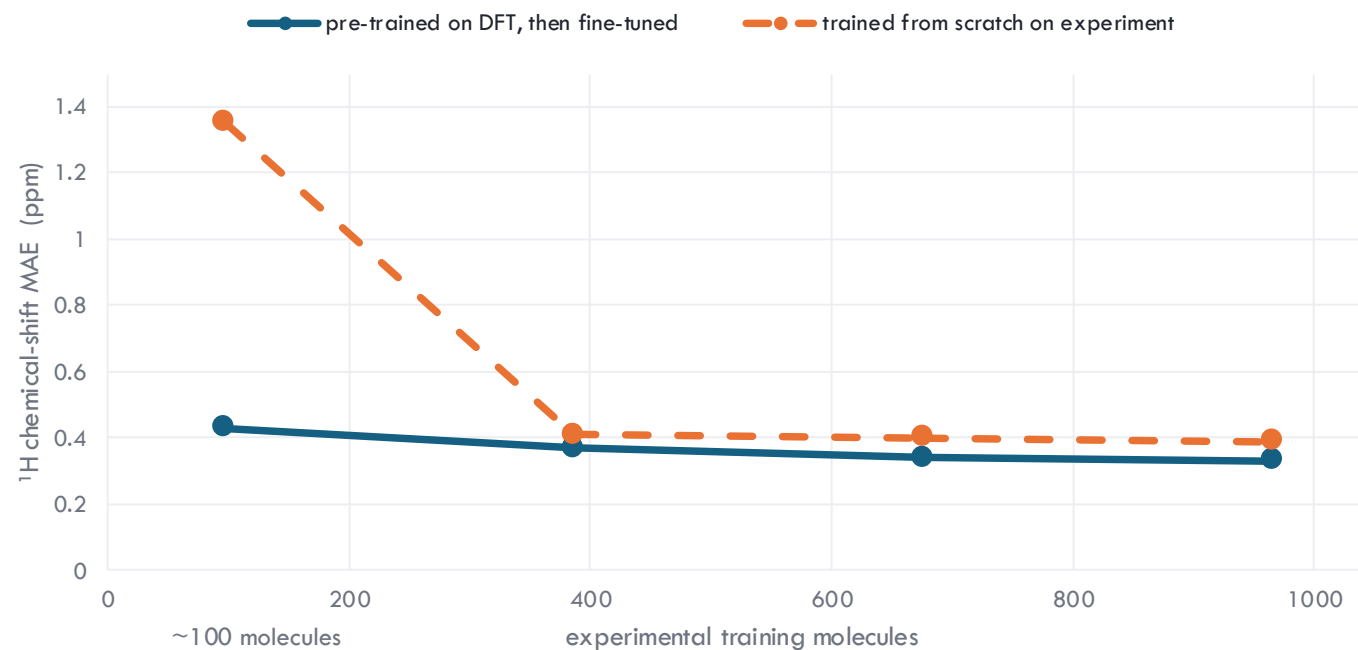
The expensive half is **amortised across the whole community**. You are buying a billion molecules' worth of chemistry for a coffee break.

Three knobs — pick by your dataset size



Always run the frozen probe first. Two minutes, no risk, and it tells you whether the representation contains anything useful for your task.

Where pre-training genuinely pays



the whole lecture in one plot

- Pre-training buys you data you do not have.
- Its value is largest when you have almost nothing.
- It decays to zero — and for ^{13}C it went negative — once you have ~1,000 labels.

Pre-training is a prior: worth 3× when you have ~100 labels, worth nothing by ~1,000, and occasionally worth less than nothing.

The same shape, in three different fields

REACTION YIELD

98

reactions are enough for a pre-trained reaction encoder to match DFT descriptors + RF

Buchwald–Hartwig HTE · Yield-BERT

INTERATOMIC POTENTIALS

3 %

of the data needed to specialise a foundation MLIP to liquid water, vs training from scratch

MatterSim · 17 M structures pre-trained

SUBLIMATION ENTHALPIES

tens

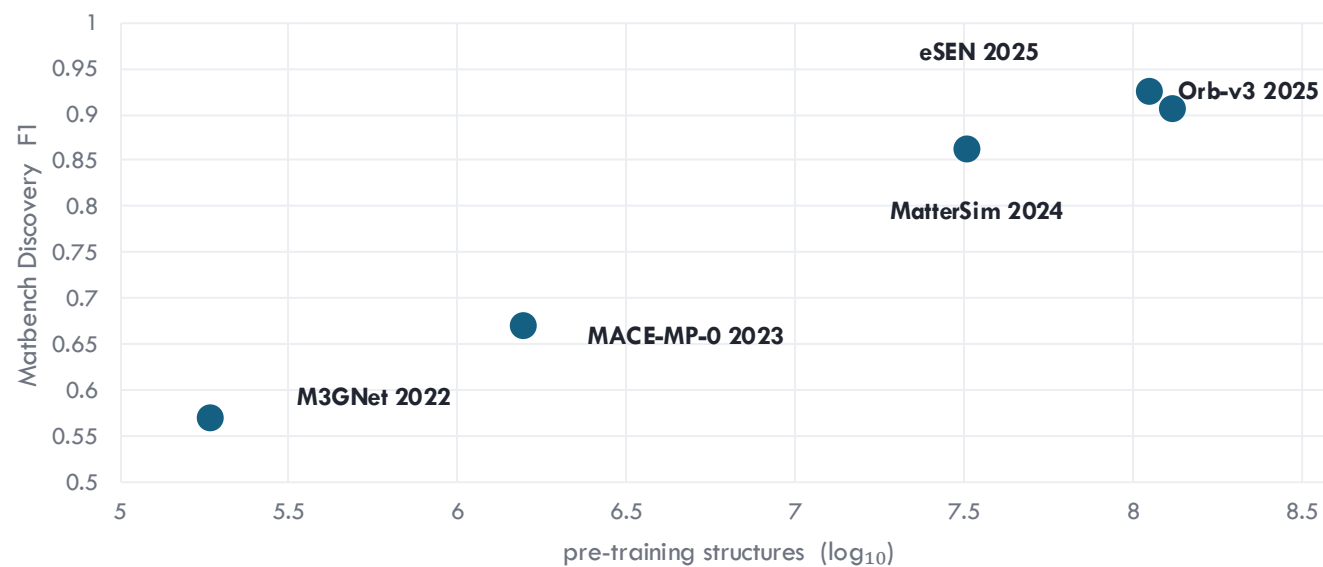
of training structures give sub-kJ/mol accuracy after fine-tuning MACE-MP-0

Kaur et al., Faraday Discuss. 2025

Different fields, different architectures, **the same shape** — the value of pre-training is concentrated where labels are most expensive.

Foundation models for atoms — the same play, at scale

~600× more training structures in three years



why chemists care

You will almost never train one of these.

You download it, then fine-tune on the 50–500 structures you can afford to compute for your own system.

Exactly the recipe from the last slide — applied to potentials instead of properties.

You will never train one of these — you will fine-tune one. Same recipe, applied to potentials instead of properties.

Matbench Discovery (matbench-discovery.materialsproject.org), values as of 11 Aug 2026 · Riebesell et al., Nat. Mach. Intell. 2025, 7, 836

06

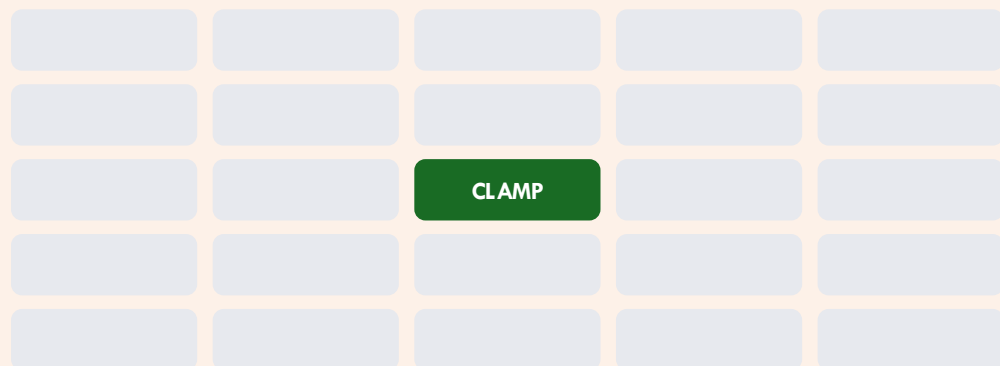
When it fails

- Negative transfer and the replication problem
- Activity cliffs
- The baseline you must always run

The uncomfortable benchmark

25 pre-trained models × 25 datasets

*re-evaluated with hierarchical Bayesian testing
instead of leaderboard means*



**one model beats the ECFP baseline significantly —
and it is itself fingerprint-based**



*nearly all neural models show
negligible or no improvement over
the baseline ECFP molecular
fingerprint.*

Prski, Adamczyk & Czech, arXiv:2508.06199 (2025)

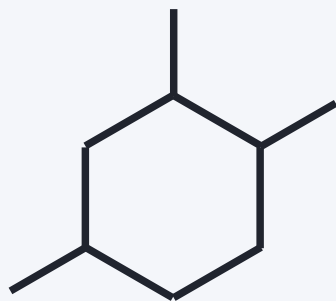
**Not a reason to give up on learned representations.
A reason to always run the ECFP + random-forest
baseline first.**

Always run ECFP + random forest first. If your model cannot beat it, you have learned something valuable for the price of two minutes.

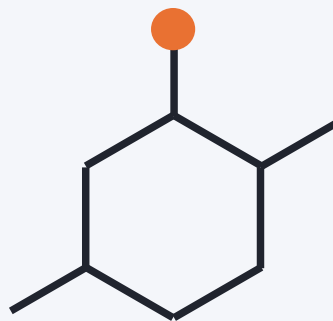
Prski, Adamczyk & Czech, arXiv:2508.06199 (2025) · Jiang et al., J. Cheminform. 2021, 13, 12 · Deng et al., Nat. Commun. 2023, 14, 6395

Activity cliffs: where smoothness is a lie

two molecules, one methyl group apart



$IC_{50} = 8 \text{ nM}$



$IC_{50} = 4,200 \text{ nM}$

Tanimoto similarity 0.95

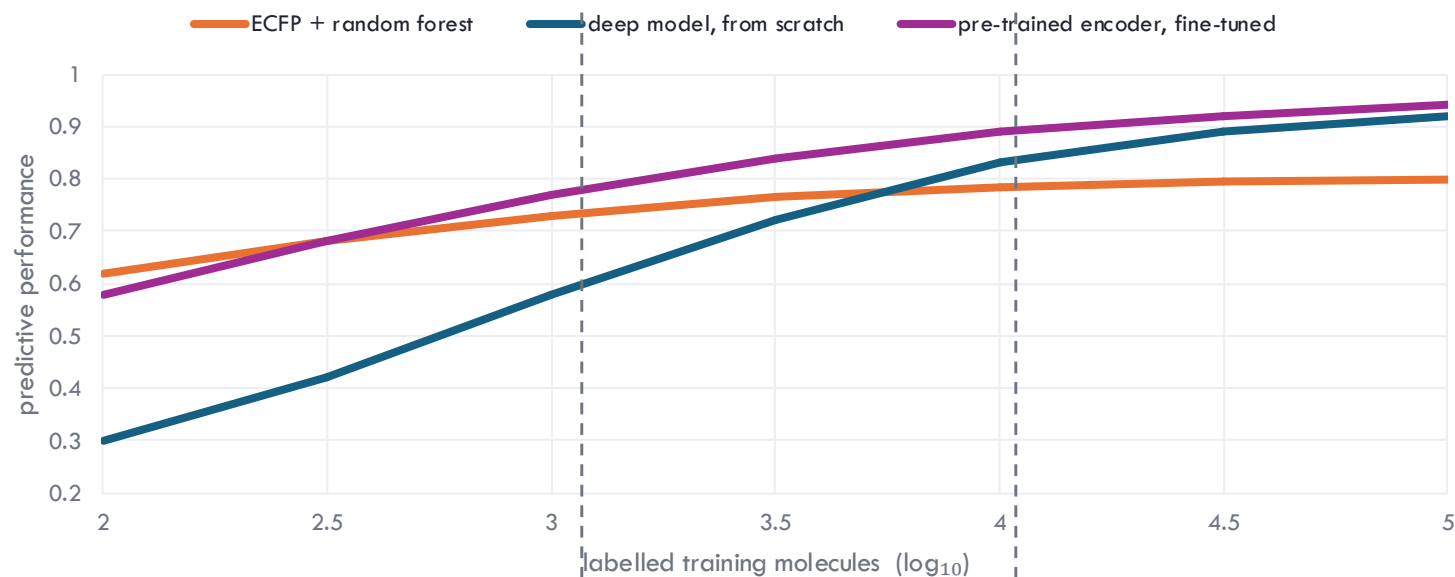
500× potency difference

what breaks

- Smooth embeddings put A and B in the same place.
- Errors on cliff compounds reach 1.4 log units — a 25× error in potency.
- Across 30 targets, SVM on ECFP was the best method overall.
- Transformers pre-trained on 10 M SMILES did not beat LSTMs pre-trained on 36 k.

Smoothness is an assumption, and cliffs violate it. **Report cliff performance separately** — overall RMSE hides exactly the compounds that matter.

So when is a learned representation worth it?



fingerprints win

pre-training earns its keep

everything converges

schematic

The shapes are real;
the exact crossover is not.

Published estimates put it
anywhere between 10^3 and 10^4
labelled molecules, and it moves
with label noise, split type and
how narrow your chemistry is.

Measure it on your own data —
that is what the tutorial does.

The shapes are robust; **the crossover is yours to measure**, and it moves with label noise, split type and how narrow your chemistry is.

When to pretrain?

How many labelled examples do you have?

< 100

Do not train a deep model.
ECFP + RF, or a Gaussian process.
Consider frozen pre-trained embeddings
as extra features.

100 – 5,000

This is where pre-training pays.
Frozen embeddings first, then partial
fine-tuning. Always report the ECFP baseline.

> 10,000

Train from scratch on your own data.
Pre-training becomes a convenience,
not a necessity.

In all three columns: run the baseline first, and split by scaffold, not at random.

Take-home messages

- **Labels are the bottleneck, not molecules.** Everything today is a way of spending unlabelled data to make up for scarce labels.
- **Unsupervised methods find real structure** — reaction classes, the periodic table — but a t-SNE plot tells you about your fingerprint, not about chemistry.
- **The VAE's decoder disappointed; its encoder changed the field.** Pretext task → keep the encoder → reuse it. That is the whole pattern.
- **Validity was the easy metric.** 100% valid is achieved by construction and says nothing about quality, novelty or synthesizability.
- **Pre-training is a prior.** Worth $\sim 3\times$ at a hundred labels, worth nothing at ten thousand, occasionally worth less than nothing.
- **Always run ECFP + random forest, and always split by scaffold.** In the largest independent replication, almost nothing beat that baseline.

Further reading

- **Chemical VAE** — Gómez-Bombarelli et al., ACS Cent. Sci. **2018**, 4, 268
- **SELFIES** — Krenn et al., Mach. Learn.: Sci. Technol. **2020**, 1, 045024 · and the critique: Skinnider, Nat. Mach. Intell. **2024**, 6, 437
- **Pre-training GNNs** — Hu et al., ICLR **2020**, arXiv:1905.12265
- **MolFormer** — Ross et al., Nat. Mach. Intell. **2022**, 4, 1256 · **MolCLR** — Wang et al., Nat. Mach. Intell. **2022**, 4, 279
- **Reaction space** — Schwaller et al., Nat. Mach. Intell. **2021**, 3, 144
- **Transfer for NMR** — Han & Choi, J. Phys. Chem. Lett. **2021**, 12, 3662
- **Do they actually help?** — Deng et al., Nat. Commun. **2023**, 14, 6395 · Praski et al., arXiv:2508.06199 (**2025**) · van Tilborg et al., JCIIM **2022**, 62, 5938
- **Evaluation guidelines** — Bender et al., Nat. Rev. Chem. **2022**, 6, 428

Questions

Learned Representations

Tutorial · 15:15 · RSC 275

Chemical space maps · frozen vs learned embeddings · the learning-curve crossover

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